OXFORD

INTERNATIONAL AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

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OXFORD AQA INTERNATIONAL AS

BIOLOGY (9610)

Unit 2 Biological systems and disease

Wednesday 17 January 2018 06:00 GMT Time allowed: 1 hour 30 minutes

A

Materials

For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 75.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Answer **all** questions in the spaces provided.

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0 1 The p53 gene is a tumour suppressor gene.

The p53 gene codes for a protein. This protein inhibits the cell cycle and helps to repair faulty DNA. A mutation in the p53 gene can cause tumour formation.

Table 1 gives the mRNA codons of 7 different amino acids.

Table 1

Amino acid	mRNA codons
Asn	AAU, AAC
Cys	UGU, UGC
Gln	CAA, CAG
Leu	CUU, CUC, CUA, CUG, UUA, UUG
Lys	AAA, AAG
Met	AUG
Phe	UUU, UUC

0 1 . 1 The mutation in the p53 gene changes one DNA triplet from AAA to AAT.

Name the amino acid coded for by the DNA triplet AAT.

Use information from **Table 1**.

[1 mark]

Do not write outside the box

0 1 . 2 Part of the DNA sequence of the p53 gene is:

T T G T T C T A C A A A A C G G T T

The mutation described in Question **01.1** occurs within this sequence.

Suggest a type of mutation that could produce this change.

Explain your answer.

[2 marks]

0 1 . 3 Why would the gene mutation produce a non-functional p53 protein?

[1 mark]

0 1 . 4 The mutation in p53 described in Question **01.1** has been found in patients with lung cancer.

Suggest **one** environmental factor that might increase the rate of mutation in the p53 gene and cause lung cancer.

[1 mark]

5

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0 2 Plants absorb mineral ions from the soil.

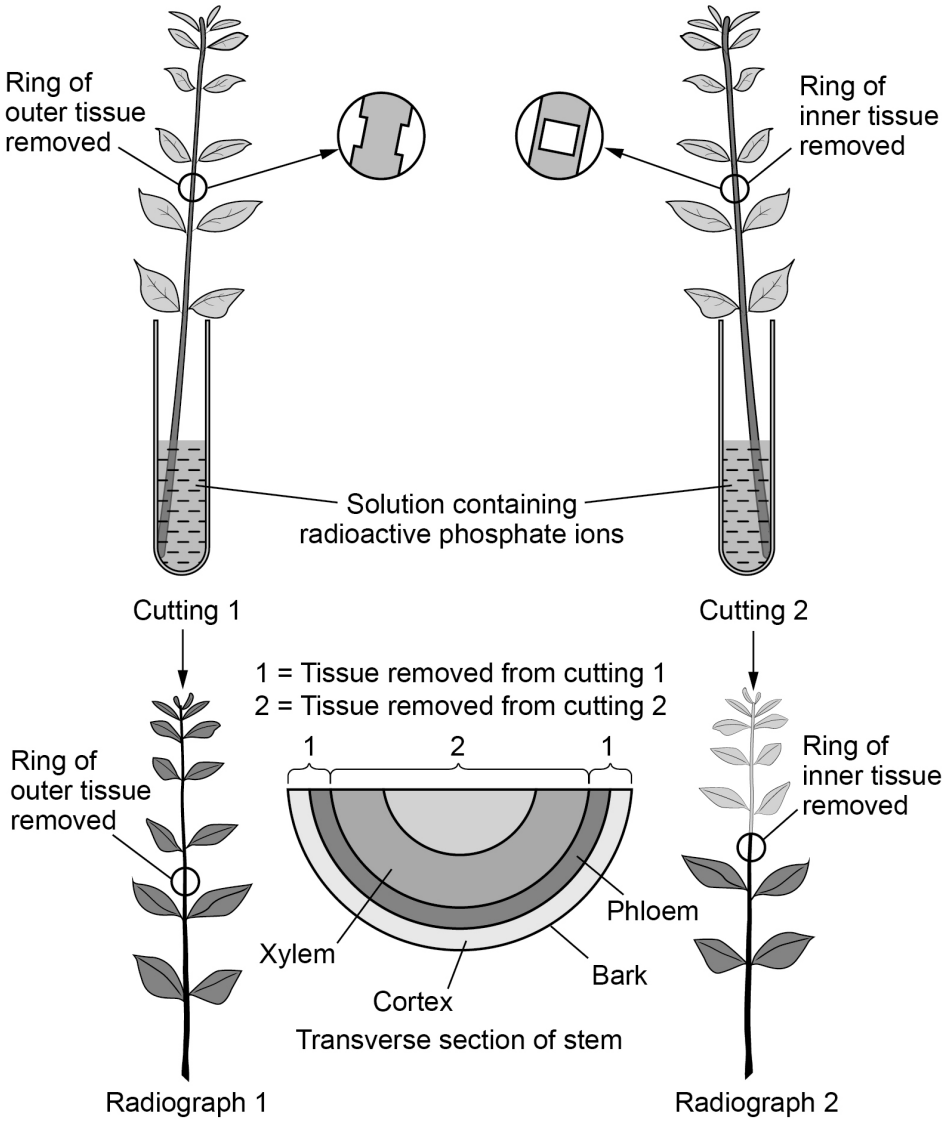
Scientists investigated the pathway taken by phosphate ions through a woody plant cutting.

The scientists:

- removed a ring of tissue, 1 cm in length, from the outside of Cutting **1**
- removed the inner tissue from 1 cm of Cutting **2**, leaving the top and bottom of the stem connected by a few strands of outer tissue
- placed each cutting in a test tube with a solution containing radioactive phosphate ions for 1 hour
- removed each cutting from its solution and placed the cutting in contact with X-ray film for several days.

Figure 1 shows the scientists' methods and the results.

Figure 1



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0 2 . 1 Which tissue transported phosphate ions up the stem of the plant cutting?

[1 mark]

Tick (✓) **one** box.

- Bark ☐
- Cortex ☐
- Phloem ☐
- Xylem ☐

0 2 . 2 Give evidence for your answer to Question **02.1**.

Include information from the results for Cutting **1** and from the results for Cutting **2** in your answer.

[2 marks]

0 2 . 3 One scientist suggested that the inner tissue from Cutting **2** may not have been completely removed.

Give evidence for this conclusion from Radiograph **2** in **Figure 1**.

[1 mark]

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0 2 . 4 The scientists then performed a second investigation with a new cutting, with tissue removed in the same way as in Cutting **2** in **Figure 1**.

In their second investigation, the scientists used radioactive sucrose solution instead of radioactive phosphate solution.

Predict the results of this investigation.

[1 mark]

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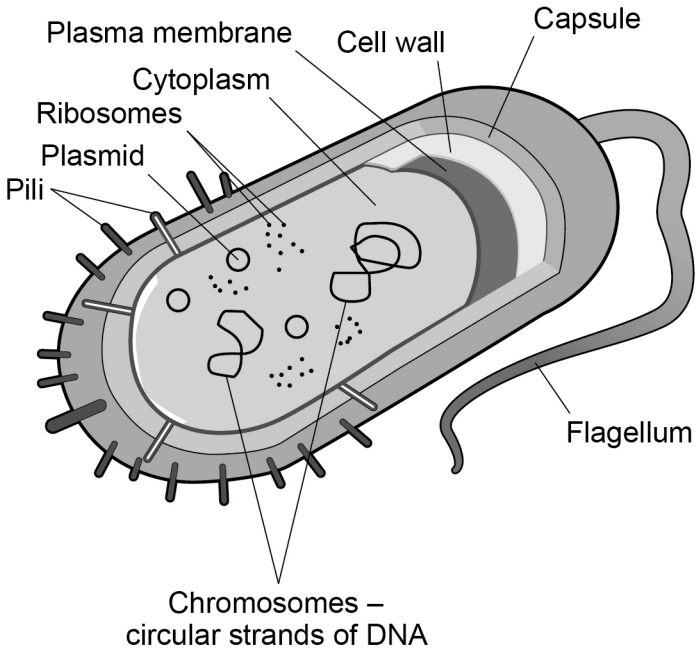
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ANSWER IN THE SPACES PROVIDED

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0 3 Cholera is a disease caused by the bacteria *Vibrio cholerae*. The disease is transmitted by drinking water containing the bacterium.

Figure 2 shows the structure of *Vibrio cholerae*.

Figure 2



0 3 . 1 In what way is the genetic material in *Vibrio cholerae* different from the genetic material found in most bacteria?

[1 mark]

Tick (✓) **one** box.

Most bacteria have RNA as the genetic material. ☐

Most bacteria have linear DNA. ☐

Most bacteria have DNA only in the mitochondria. ☐

Most bacteria have one circular chromosome. ☐

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0 3 . 2 *Vibrio cholerae* lives in water.

Describe how **one** feature, shown in **Figure 2**, is an adaptation to living in water.

[1 mark]

Vibrio cholerae produces a toxin. This toxin causes disease.

The toxin molecule has two types of subunit, **A** and **B**. Subunit **B** binds to the cell membranes of the epithelial cells lining the human intestine. Subunit **A** then enters the epithelial cells and activates chloride ion channels in the cell membranes.

0 3 . 3 Explain how the cholera toxin can cause severe diarrhoea in a person infected by *Vibrio cholerae*.

[3 marks]

Scientists have produced a vaccine against cholera. The vaccine contains subunit **B** of the cholera toxin.

0 3 . 4 Why did the scientists include only subunit **B**, and **not** the whole toxin molecule, in the vaccine?

[1 mark]

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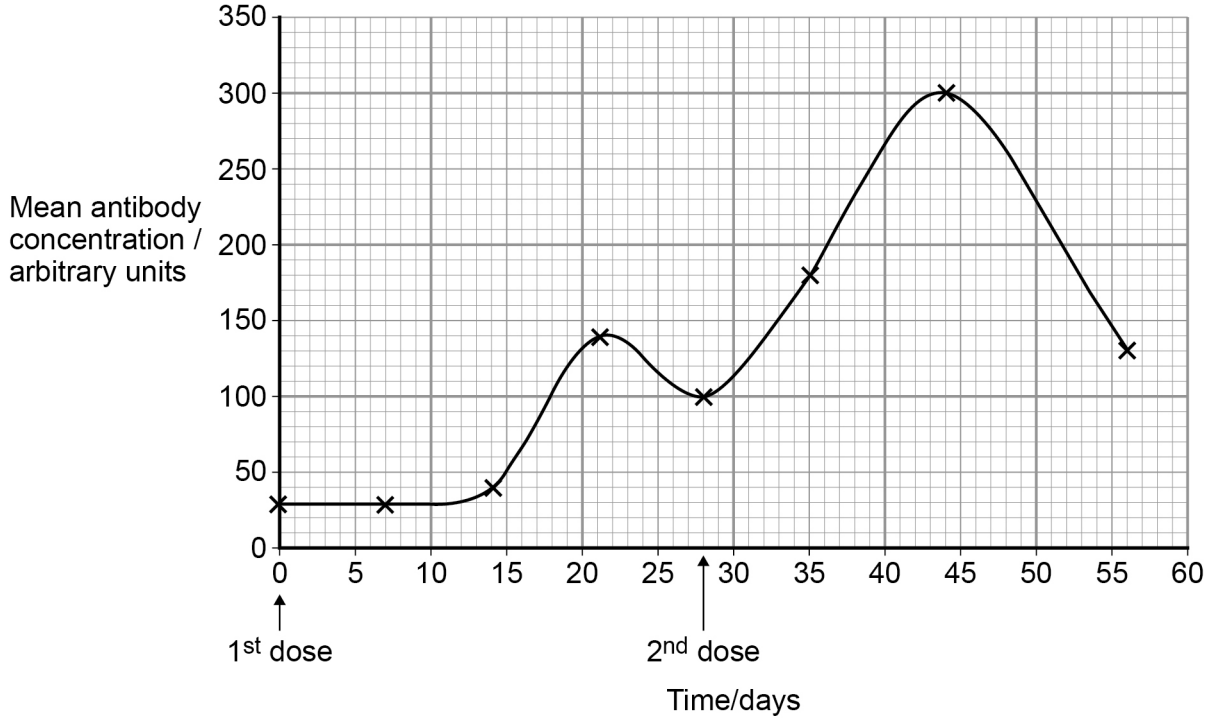
The scientists tested the vaccine on human volunteers.

They gave each volunteer two doses of subunit **B**, the first dose at the start of the investigation and the second dose 28 days later.

The scientists measured the concentration of antibodies specific to subunit **B** in each volunteer's blood plasma.

Figure 3 shows the results.

Figure 3



0 3 . 5 The second dose of Subunit **B** was more effective than the first dose in stimulating antibody production.

How many times higher is the peak antibody concentration after the second dose than the peak antibody concentration after the first dose?

[1 mark]

_____ times

Do not write outside the box

0 3 . 6 The concentration of antibody is higher after the second dose than the first dose.

Give **one** other difference in the response to the second dose.

[1 mark]

0 3 . 7 Describe what happens in a person's body in response to the first dose of subunit **B**.

[3 marks]

0 3 . 8 Give the reason why the second dose of subunit **B** has a greater effect than the first dose.

[1 mark]

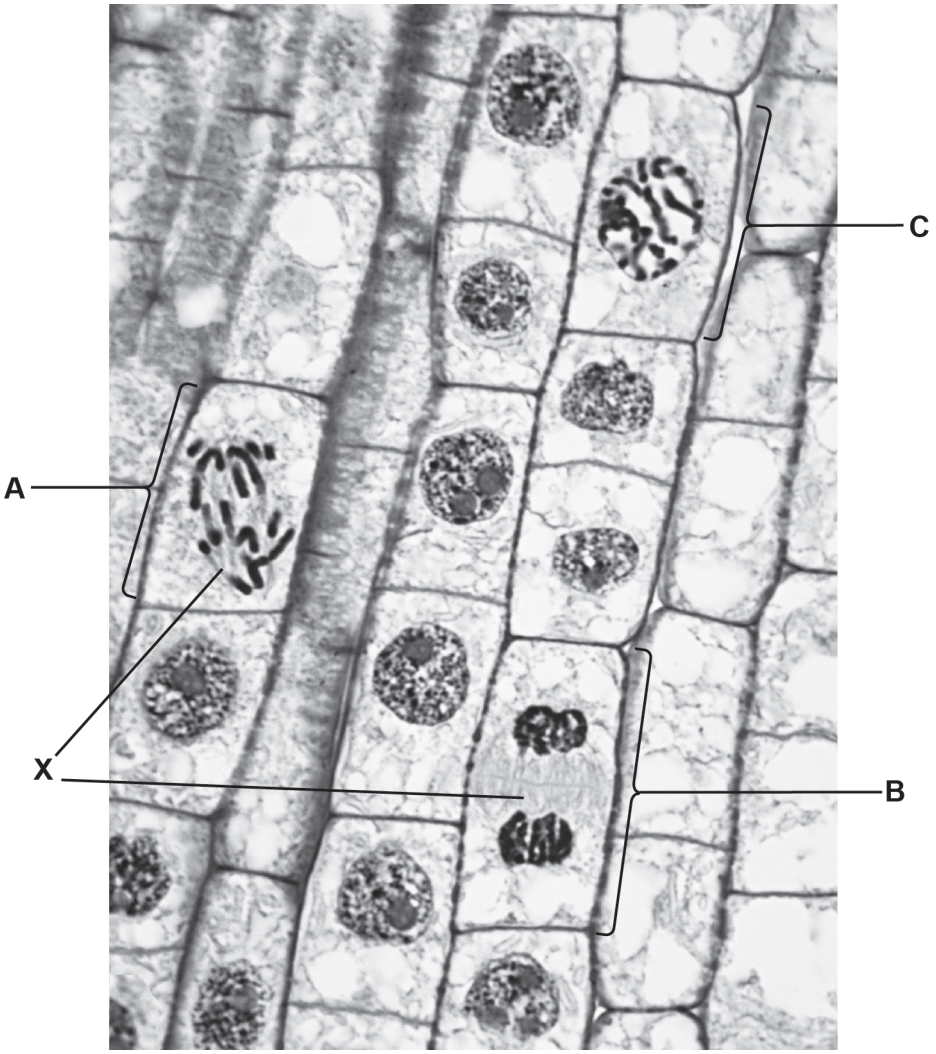
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0 4 **Figure 4** shows some cells from the root tip of an onion plant.

Figure 4



0 4 . 1 Cells **A**, **B** and **C** are in different stages of mitosis.

Give the name of the mitotic stage for each cell.

[3 marks]

Cell **A** _____

Cell **B** _____

Cell **C** _____

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0 4 . 2 Structure **X** is visible in each of the cells **A** and **B**.

Name structure **X** and give its function.

[2 marks]

Name _____

Function _____

_____

_____

A student used information from **Figure 4** to calculate that the mitotic index of the root tissue was 0.20

When calculating the mitotic index, the student did not include every cell visible in **Figure 4**.

0 4 . 3 How many cells did the student count?

[2 marks]

Number of cells counted = _____

0 4 . 4 Suggest why the student counted the number of cells you gave in your answer to Question **04.3**.

[1 mark]

_____

_____

_____

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0 4 . 5 The student's value of 0.20 for the mitotic index of the root tissue is probably not accurate.

Describe what the student could do to obtain a more accurate mitotic index for the root tissue.

Give a reason for your answer.

[2 marks]

10

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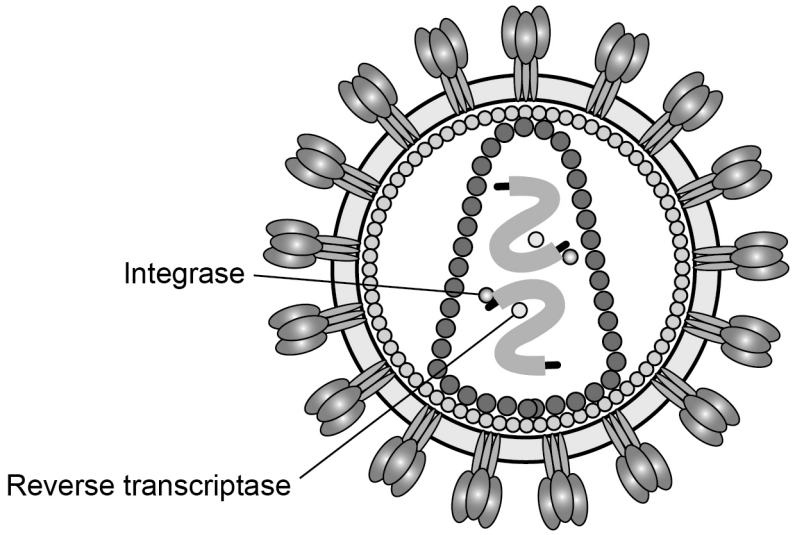
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0 5 **Figure 5** shows the structure of the human immunodeficiency virus (HIV).

Figure 5



0 5 . 1 On **Figure 5**, label the following structures.

- Capsid
- Glycoprotein

[2 marks]

0 5 . 2 HIV contains the enzymes integrase and reverse transcriptase.

A human CD4 helper T-cell contains the enzyme DNA polymerase.

Describe the function of each of these enzymes in the infection of a CD4 helper T-cell with HIV.

[3 marks]

Integrase

Reverse transcriptase

DNA polymerase

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The genetic material in HIV mutates at a very high rate. Mutations in the genes coding for integrase and reverse transcriptase are very common.

Many of the antiviral drugs used for treatment of HIV-infected patients are inhibitors of **either** integrase **or** reverse transcriptase.

Treatment with a single drug often results in the development of resistance in the virus and the patient becomes ill.

Treatment with two drugs at the same time, each drug targeting a different enzyme, does not result in resistance. The patient stays healthy.

0 5 . 3 Suggest an explanation for the different effects of treatment with a single antiviral drug and treatment with two drugs at the same time.

[4 marks]

9

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0 6 During exercise, the heart beats faster and more forcefully.

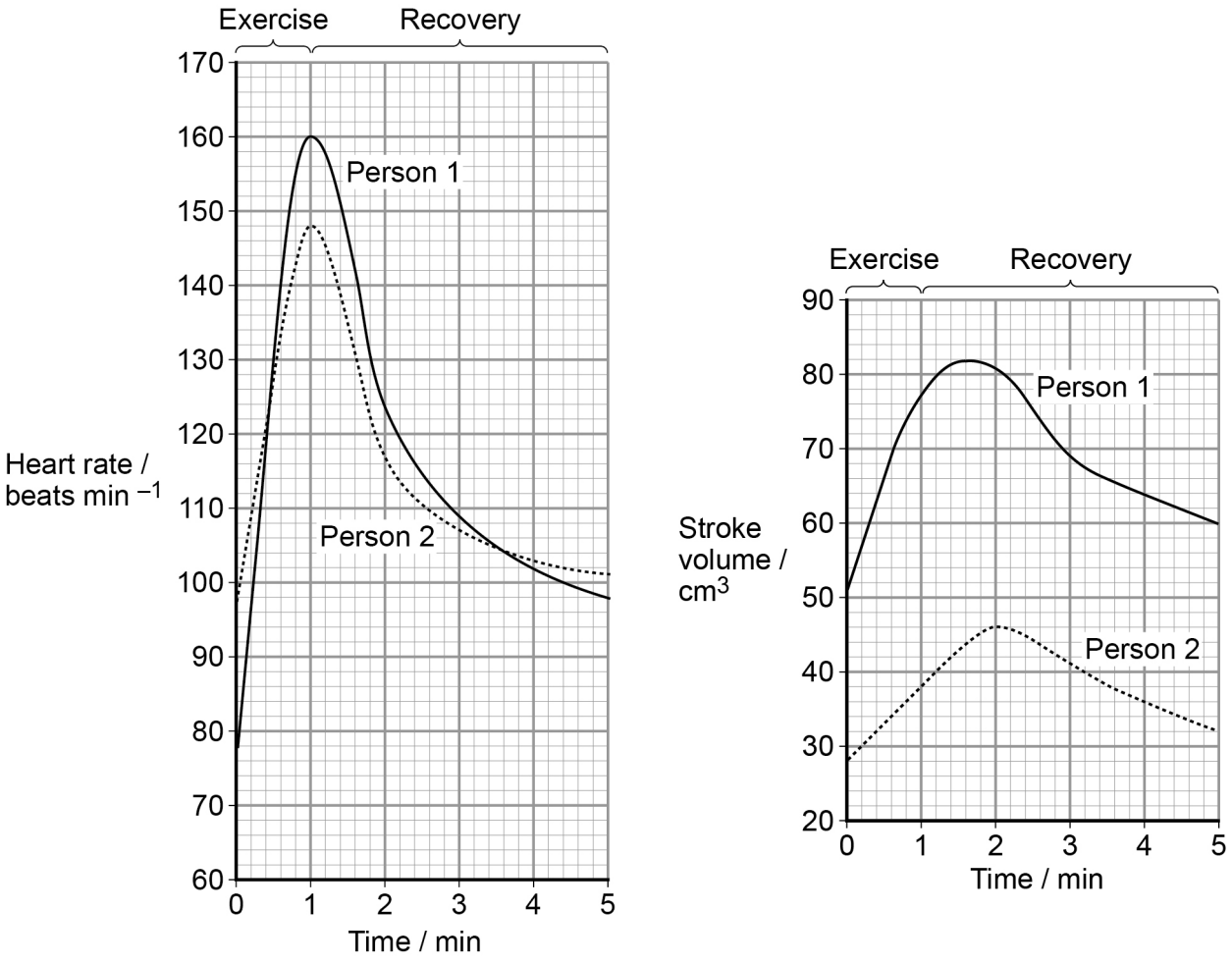
In an investigation, **Person 1** and **Person 2** each ran on a treadmill set at a speed of 14 km h^{-1} for one minute.

Scientists measured the heart rate and stroke volume of each person:

- at rest before the run
- at the end of the run
- at 1-minute intervals during recovery after the run.

Figure 6 shows the scientists' results.

Figure 6



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0 6 . 1 Each person ran on a treadmill in a sports laboratory.

Explain **two** advantages of monitoring the effect of exercise on people running on treadmills in a sports laboratory rather than running on an outdoor athletics track.

[2 marks]

1 _____

_____

_____

2 _____

_____

_____

0 6 . 2 **Person 1**'s cardiac output at one minute was 12.4 $dm^3\ min^{-1}$.

Calculate **Person 2**'s cardiac output in $dm^3\ min^{-1}$ at one minute.

Use information from **Figure 6** and give your answer to an **appropriate** level of number of significant figures.

[2 marks]

Person 2's cardiac output at one minute = _____ $dm^3\ min^{-1}$

Do not write outside the box

0 6 . 3 Use information from **Figure 6**, and your own knowledge of the functioning of the heart, to explain the **main** reason why **Person 1** had a higher cardiac output than **Person 2**.

[2 marks]

0 6 . 4 **Person 2** smoked 20 cigarettes a day and had a diet high in fat.

These factors lowered the efficiency of **Person 2**'s heart.

Explain how.

[4 marks]

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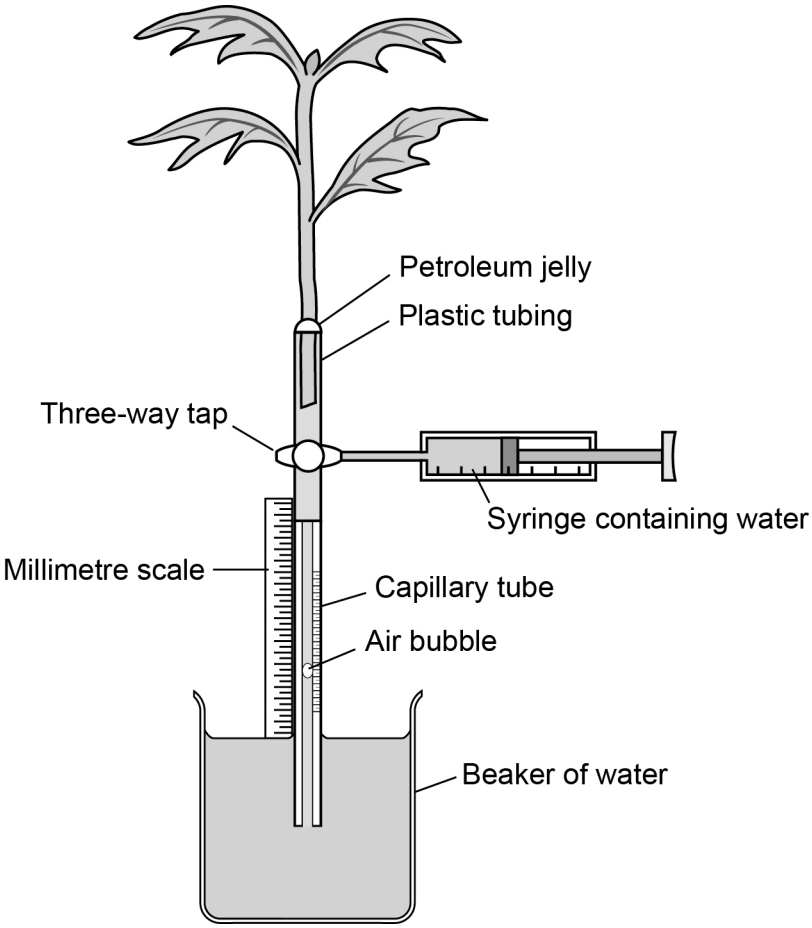
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0 7 A potometer is used for measuring the rate of water uptake by a plant cutting.

Some students set up a potometer as shown in **Figure 7**.

Figure 7



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0 7 . 1 Describe the method the students should use to assemble the apparatus as shown in **Figure 7**.

Give reasons for any precautions the students should take.

[4 marks]

Question 7 continues on the next page

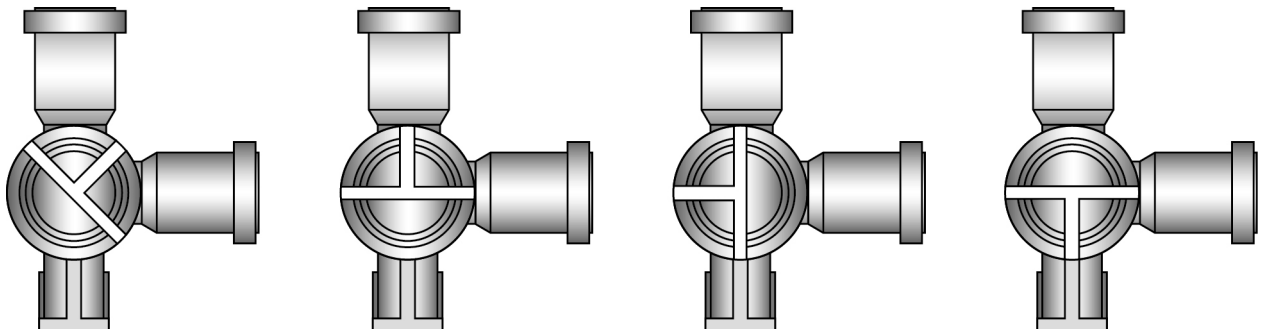
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The students recorded the position of the air bubble on the millimetre scale at 1-minute intervals for 5 minutes.

At the end of the 5 minutes, the students used the syringe to return the air bubble to the lower end of the capillary tube.

Figure 8 shows four possible positions of the three-way tap, **A**, **B**, **C**, and **D**.

Figure 8



0 7 . 2 Which position, **A**, **B**, **C** or **D**, should the students use when returning the air bubble to the lower end of the capillary tube?

[1 mark]

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Table 2 shows the students' results.

Table 2

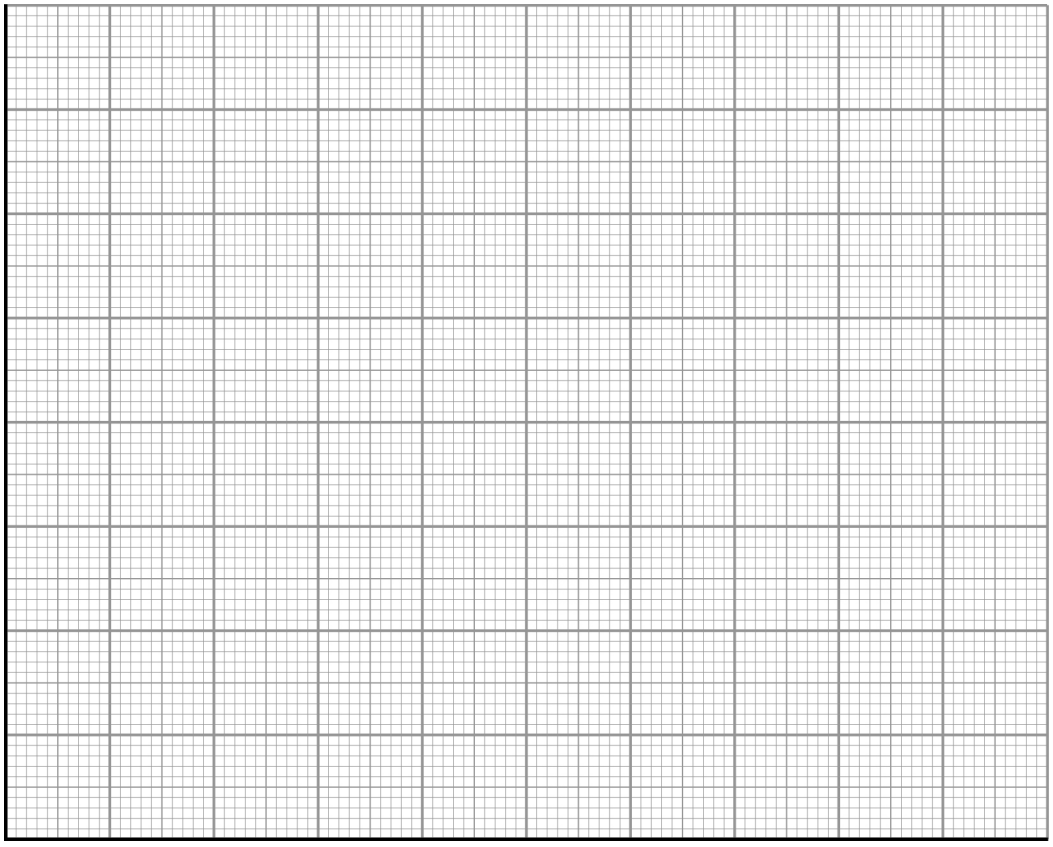
Time / min	Position of air bubble / mm
0	12
1	66
2	113
3	165
4	220
5	272

0 7 . 3 On **Figure 9**, draw a graph of the students' results.

Draw a suitable line of best fit.

[3 marks]

Figure 9



Turn over ►

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0 7 . 4 Determine the gradient of the line on your graph.

Give your answer in mm min^{-1}.

[1 mark]

Gradient = _____ mm min^{-1}

0 7 . 5 The internal diameter of the capillary tube in **Figure 7** is 1 mm. The density of water is 1.0 g cm^{-3}.

Use this information, and your answer to Question **07.4**, to calculate the rate of water uptake by the students' plant cutting in mg h^{-1}.

Give your answer in standard form, correct to 3 significant figures.

Use $\pi = 3.14$

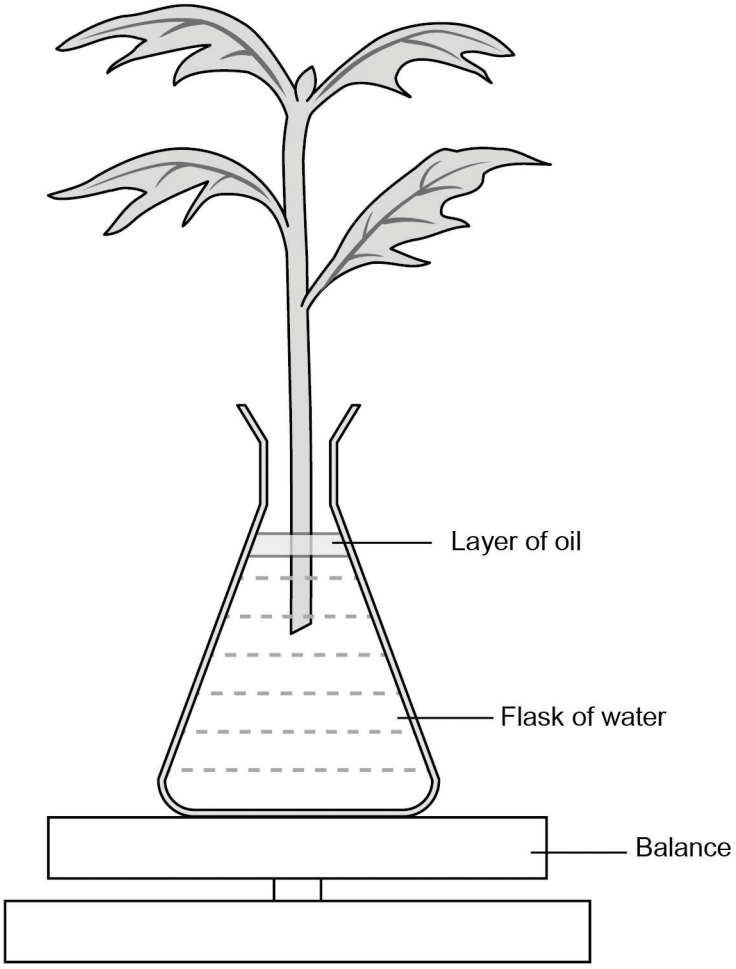
[2 marks]

Rate of water uptake = _____ mg h^{-1}

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The students removed the plant cutting from the potometer and placed it in the apparatus shown in **Figure 10**.

Figure 10



0 7 . 6 The layer of oil on top of the water prevented evaporation directly from the water in the flask.

State why is this important.

[1 mark]

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0 7 . 7 The students recorded the reading on the balance at the start and after 24 hours. From this, they calculated the rate of water loss from the cutting.

The rate of water loss measured using this apparatus was similar to the rate of water uptake measured by the students using the potometer.

Explain why.

[2 marks]

14

Turn over for the next question

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0 8 The food we eat is digested and absorbed in the digestive system.

0 8 . 1 Describe how proteins are digested in the human digestive system.

[5 marks]

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0 8 . 2 Explain how each of the following are involved in the absorption of the products of protein digestion by the ileum.

- Active transport
- Co-transport with sodium ions (Na^+)
- Diffusion

[5 marks]

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END OF QUESTIONS

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